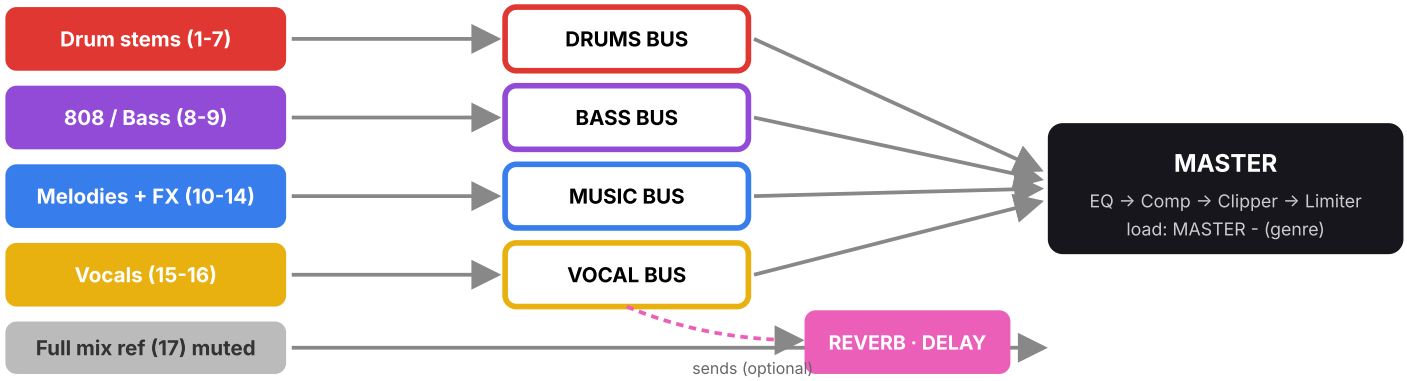
Mixing Cheat Sheet

Stems + vocals in FL Studio 21 · your Stem Mixer template, master presets, delay presets and 18 vocal presets

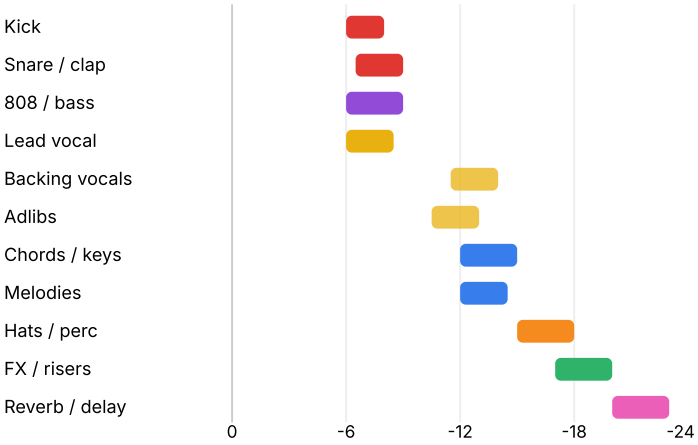
1 · How your mix flows



Balance the song with the **4 bus faders**. Use single tracks only to fix one sound. Vocal presets go on tracks 15-16 (or any track).

2 · Starting levels (peak meter, before the master chain)

Where each part should peak (dBFS)



The 3 rules that matter

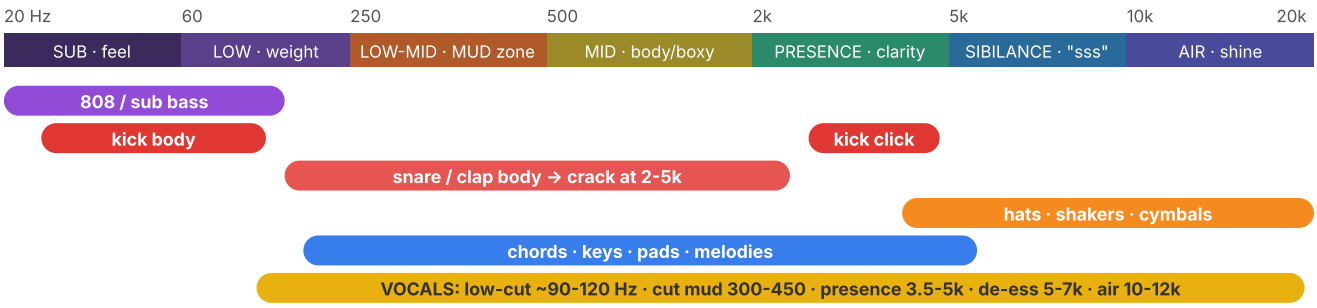
1. **Master peaks around -6 to -3 dB** before the mastering chain. Never in the red while mixing.
2. **Lead vocal sits 3-6 dB above the loudest instrument**. It's the star.
3. **Backing = 6-9 dB under the lead. Adlibs = 4-6 dB under.** Doubles: 4-6 dB under.

Balancing order

1. Kick alone → set it.
2. Add 808 / bass until it feels equal to the kick by ear.
3. Snare/clap, then hats/perc.
4. Lead vocal on top.
5. Chords → melodies → FX, tucked under the vocal.
6. Switch to mono for 10 s: does the vocal and 808 stay solid?

Bars are starting points from common mix guides. Your ears decide. Level-match before judging "better".

3 · Frequency map: who lives where



Two things fighting in the same zone = mud. Fix by **cutting one** of them there (EQ), or **ducking** one under the other (sidechain), not by turning both up.

4 · Your 24 vocal presets: pick in 3 steps

1 · GENRE
Hip Hop = rap
Afrobeats = sung, bright
RnB = sung, smooth
House = sung, big space

→

2 · VOICE
Male = deeper voice
Female = higher voice

→

3 · ROLE
Lead = main vocal, upfront
Backing = harmonies, doubles
Adlibs = "yeah!", chants, shouts

Load one: Browser (left) → **Mixer presets** → drag e.g. **VOX - Afrobeats - Female Lead** onto your vocal's mixer track. Or right-click the track name → **File** → pick it. The full 8-plugin chain loads in one go.

What's inside every vocal preset (in order)

1 Pitcher	2 EQ clean	3 De-esser	4 Comp FAST	5 Comp SMOOTH	6 EQ tone	7 Delay	8 Reverb
OFF (tuning)	low-cut, mud	Maximus	catches peaks	evens it out	presence, air	tempo-synced	space

Order from research: clean EQ → de-ess → compress in 2 stages → tone EQ → time effects (Ghostnote, Rys Up Audio, Audient)

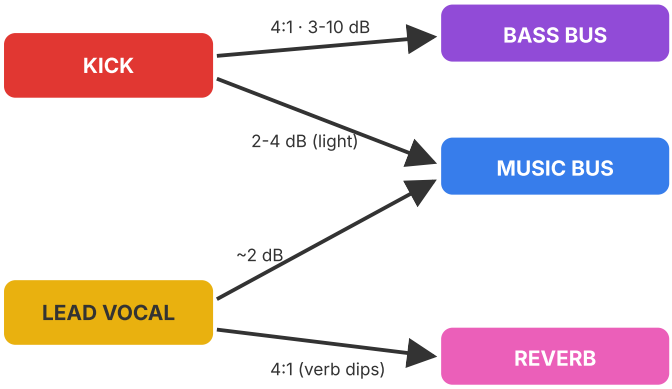
	Lead	Backing	Adlibs
Low-cut (M / F)	90 / 120 Hz	170 / 200 Hz: thinner, so it doesn't crowd the lead	250 / 280 Hz: thin, "in the back"
Mud cut (M 300 · F 450 Hz)	-2.5 dB	-4 dB	-3 dB
Presence (M 3.5k · F 5k)	+2 dB	**-2 dB** (lead wins)	+2.5 dB (cuts through)
Top end	Air shelf +2.5 to +3.5 dB (M 10k · F 12k)	Rolled off above 11 kHz	Air +1.5 dB
Compression	Genre ratio (below)	4:1 fast comp: more even, sits back	
Delay / reverb	Low: delay 15%, small verb	More: delay 18%, longer verb	Most: delay 25%, medium verb

	Hip Hop	Afrobeats	RnB	House
Fast comp	4:1, 1.5 ms attack: tight, aggressive	3:1, 5 ms attack	3:1, 15 ms attack: keeps the breath	4:1, 3 ms, fast 80 ms release: steady on the beat
Smooth comp	2.5:1, 25 ms	2:1, 30 ms	2:1, 30 ms, 250 ms release	2:1, 30 ms, 200 ms
Reverb	~0.9 s, small: dry and upfront	~1.2 s plate: bright	~2.2 s: lush	~1.8 s, bigger room: airy
Delay	1/4 throws · adlibs 1/8	1/4 last-word throw · backing 1/8 dotted	1/8 dotted · backing 1/4 dotted	Groove 1/8 dotted · backing Wash 1/4

Tweak these 3 things per song

1. **Compressor THRESHOLD** (slots 4 and 5): sing/rap the loudest line and aim for **3-6 dB** of turning down on slot 4 and **1-3 dB** on slot 5. If your vocal is quiet, lower the threshold. If loud, raise it.
2. **De-esser**: still hissy on "s"? Open Maximus and raise the mid band's compression. Lisping? Lower it.
3. **Pitcher** (slot 1, OFF): for sung vocals, switch it ON and set the song's key. Leave it off for rap.

5 · Sidechain map: who ducks under whom



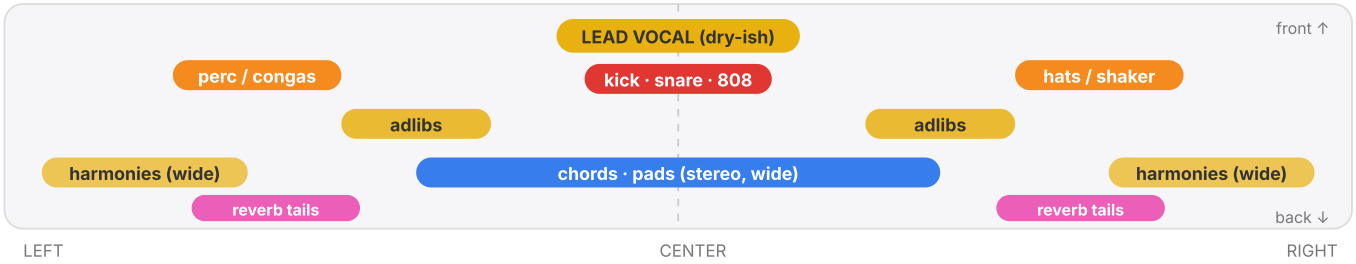
How to link (every time):

1. Click the **trigger** track (KICK or LEAD VOCAL).
2. Right-click the small arrow under the **target** → *Sidechain to this track*.
3. Target's **Fruity Limiter**: CEIL up → **COMP** → right-click SIDECHAIN → pick trigger → lower threshold.

Duck amount	HH	Afro	RnB	House
Kick → Bass	7-10 dB	4-6 dB	3-5 dB	6-10 dB
Kick → Music	2-3	2-4	0-2	3-6 (pump)
Vocal → Music	~2	~2	~2	~2

No kick stem? Use **DRUMS LOW** as the trigger. Clicky? Raise REL.

6 · Stereo stage: where things sit (seen from above)



Center: kick, snare, 808/bass (mono), lead vocal. **Sides:** perc, hats, harmonies, guitar, arps. **Front ↔ back** = less ↔ more reverb and high-end. Doubles: 30-50% L/R, harmonies 70-100% L/R (Rys Up Audio).

7 · "It sounds..." → fix it

Problem	Try this (in order)
Muddy / cloudy	Cut 250-450 Hz on chords/pads/vocal (-2 to -4 dB) · raise low-cuts on everything that isn't kick or bass · less reverb, and low-cut the reverb at 300 Hz
Vocal buried	Raise the vocal fader first · +1-2 dB presence (3-5k) · Lead vocal → Music bus sidechain (~2 dB) · cut 2-4k a little on chords/melodies
Vocal harsh / "sss"	More de-essing (Maximus) · less presence/air boost · darker reverb (high cut 6-7k)
808 disappears on phone	Fruity Soft Clipper on the 808 for harmonics · check kick vs 808 sidechain · don't low-cut the 808
Kick and 808 fight	Kick → Bass bus sidechain · or the 808 EQ trick (Peak Controller on kick, inverted link to an EQ band under 200 Hz on the 808)
Mix sounds small / narrow	Pan perc, hats and harmonies out · stereo chords/pads · keep low end centered
Vocal sounds far away	Less reverb/delay wet · shorter reverb · more pre-delay (30-40 ms) so the voice comes first
Pumping / breathing	Less gain reduction on the master compressor · slower release on the sidechain
Distorts / red on master	Pull the 4 bus faders down, not the master · master peaks around -6 dB before limiting

8 · Every song: the 10-step routine

1. File → New, drag stems in
2. Run Stem Mixer
3. Vocal presets on 15 / 16
4. Gain stage: master -6 dB
5. Balance kick, bass, vox
6. Sidechain kick → bass
7. Fix problems (section 7)
8. Load MASTER preset
9. Limiter GAIN to LUFS target
10. Compare with FULL MIX (17)

Your presets	Where in FL	What
VOX - (genre) - (voice) (role) × 18	Browser → Mixer presets	Full vocal chain for one track
MASTER - Hip Hop / Afrobeats / RnB / House	Browser → Mixer presets	Mastering chain for the Master track
HIP HOP / AFROBEATS / RnB / HOUSE delays × 7	Fruity Delay 3 → Presets	Echo only (dry 0%), for a DELAY send track
Stem Mixer script	Options → MIDI Settings → Controller type	Builds and routes the 23-track mixer

Sources: Ghostnote (rap vocal chain; de-essing in FL Studio) · Dan Murtagh (vocal EQ cheat sheet) · Rys Up Audio (R&B vocals, hip-hop vocals) · Antares (Afrobeats vocal production) · Audient (basic vocal chain) · Adrian Milea & Modern Mixing (background vocals) · Avid (vocal EQ) · Mix Elite (sidechain in FL Studio) · Mastering-France / Venia (loudness). All numbers are starting points. Trust your ears and compare at equal loudness.